

Topic # Dimension

Definitions

- The dimension of a finite-dimensional vector space V , is defined to be the number of vectors in a basis for V .
- it is denoted by $\dim(V)$.
- The zero vector space has dimension zero.

e.g.

for example, $\dim(\mathbb{R}^n) = n$,
 $\dim(P_n) = n+1$ and $\dim(M_{mn}(\mathbb{F})) = mn$

Dimension of Span(S):

The dimension of the space spanned by a linearly independent set of vectors is equal to the number of vectors in that set.

$$\dim[\text{Span}\{v_1, v_2, \dots, v_r\}] = r$$

Theorems

Let W be a subspace of finite dimensional vector space V . Any linearly independent set

w can be expanded to a basis for w and

$$\dim w \leq \dim V$$

$$w = V \text{ iff } \dim(w) = \dim(V)$$

Theorem:-

- The dimension of $\text{NULL } A$ is the number of free variables in $AX=0$
- The dimension of $\text{Col } A$ is the number of pivot columns in A .

$$\dim(R^n) = n, \dim(P^n) = n+1$$

Basis extension theorem:-

$\Rightarrow$ Let $V(F)$ be a finite dimensional vector space and $S = \{v_1, v_2, \dots, v_k\}$ a linearly independent subset of V . Then either S itself is a basis of V or S can be extended to form a basis of V .

Theorem: Fundamental matrix space
The row space and column space of any matrix have the same dimension.

Rank and Nullity:

$\Rightarrow$ Let A be any matrix.

- o The common dimension of $\text{row } A$ and $\text{col } A$ is called rank of A , denoted by $\text{rank}(A)$.
- o The dimension of $\text{Nul } A$ is called the nullity of A , denoted by $\text{nullity}(A)$.

Theorem: Rank of transpose matrix:

Let A be any matrix. Then

$$\text{rank}(A) = \text{rank}(A^t)$$

Dimension Theorem for matrices:

- o Let A be $m \times n$ matrix.

Then

$$\text{rank}(A) + \text{nullity}(A) = n$$

$$\text{rank}(A) \leq \min(m, n)$$

Theorem

- Let A be $m \times n$ matrix. Then
- $\text{rank}(A) =$ the number of leading variables in the solution of $Ax=0$
 - $\text{nulity}(A) =$ the number of parameters in the general solution of $Ax=0$

- a) Find the number of parameters in the general solution of $Ax=0$ if A is a 5×7 matrix of rank 3.

$$\therefore \text{Nulity}(A) = n - \text{rank}(A) \\ = 7 - 3 = 4$$

Thus, there are four parameters.

- b) Find the rank 5×7 matrix A for which $Ax=0$ has a two-dimensional solution space?

The matrix A has nulity 2, so

$$\text{rank}(A) = n - \text{nulity}(A) = 7 - 2 = 5$$

What are the fundamental basis for a matrix?

There are six vector spaces associated with a matrix A and its transpose A^t :

- Row space of $A: C(A) = r$
- Column space of $A: R(A) = r$
- Null space of $A: (n-r)$
- Row space of A^t
- Column space of A^t
- Null space of $A^t: (m-r)$

n - no of vectors
 m - no of elements

Theorem: -

if A is any matrix, then
 $\text{rank}(A) = \text{rank}(A^t)$

$$\begin{aligned}\text{rank}(A) &= \dim(\text{Row space of } A) \\ &= \dim(\text{Column space of } A^t) = \text{rank}(A^t)\end{aligned}$$

Dimension of sum of subspaces

Let W_1, W_2 are subspaces of finite dimensional vector space $V(F)$

$$\begin{aligned}\text{Then } \dim(W_1 + W_2) &= \dim(W_1) + \dim(W_2) \\ &\quad - \dim(W_1 \cap W_2)\end{aligned}$$

Question: Find the dimension and a basis for the four fundamental subspaces for

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

Solution:

$$m = 3$$

$$n = 4$$

$$\sim R_3 \rightarrow R_3 - R_1 \begin{bmatrix} \text{pivot} \\ \textcircled{1} & 2 & 0 & 1 \\ 0 & \textcircled{1} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(A) = 2$$

$$\therefore r = 2$$

Basis of column space A is

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right\}$$

Basis for Nullspace

$$\text{Echelon form of } A_2 \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Null space of A is the set of all solutions such that $Ax=0$

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

x_1 & x_2 are pivot variables

$$x_1 + 2x_2 + x_4 = 0$$

$$x_2 + x_3 = 0$$

$$\Rightarrow x_2 = -x_3$$

$$\text{Now } x_1 = -2x_2 - x_4$$

$$= -2(-x_3) - x_4$$

$$x_1 = 2x_3 - x_4$$

$$\therefore X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2x_3 - x_4 \\ -x_3 \\ x_3 \\ x_4 \end{bmatrix}$$

$$X = x_3 \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Basis for the null space:

$$\left\{ \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Dimension of null space of $A = n - r$
 $\therefore 4 - 2 = 2$

Given $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$

Then

$$A^t = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{array}{l} R_4 \rightarrow R_4 - R_1 \\ R_2 = R_2 - 2R_1 \end{array} \quad \begin{array}{c} \text{Pivot} \\ \text{Pivot} \end{array} \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{Rank} = 2$$

$\Rightarrow$ Dimension of row space of A is $C(A^t) = 2$

Now, first and second column contains the pivot.

$\therefore$ Basis for row space of A is the basis for column space of A^t . i.e.

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\} \dots$$

Question#:- Find the null-space of the matrix.

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix}$$

Here,

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix}_{3 \times 4} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}_{4 \times 1} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_{3 \times 1}$$

Augmented matrix:-

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 4 & 0 \\ 4 & 3 & 2 & 1 & 0 \end{array} \right]$$

$$\begin{array}{l} \sim R_2 - R_1 \\ \sim R_3 - 4R_1 \end{array} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & -1 & -2 & -3 & 0 \end{array} \right]$$

$$\begin{array}{l} \sim R_1 - R_2 \\ \sim R_3 + R_1 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

which is RREF-form

$$x_1 + 0x_2 - x_3 - 2x_4 = 0$$

$$0x_1 + x_2 - 2x_3 + 3x_4 = 0$$

$$\Rightarrow x_1 = x_3 + 2x_4$$

$$\Rightarrow x_2 = -2x_3 - 3x_4$$

Now,

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_3 + 2x_4 \\ -2x_3 - 3x_4 \\ x_3 \\ x_4 \end{bmatrix}$$

$$N(A) = \left\{ x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\text{Basis of } N(A) = \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Standard basis: $P_3 = (1, x, x^2, x^3)$

$$V_2(R) \text{ or } R^2 = \{(1, 0), (0, 1)\}$$

$$R^3 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$R^4 = \{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)\}$$

$$R^n = \{(1, 0, 0, \dots, 0), (0, 1, 0, \dots, 0), (0, 0, 1, \dots, 0), \dots, (0, 0, 0, \dots, 1)\}$$

$$= \{e_1, e_2, \dots, e_n\}$$